docs/references/automation/catzc-azure.md

`Catzc.Azure`

The Azure identity module. It owns the *model* behind every Azure deployment — the tenants, customers, environments, and subscriptions the platform deploys into, and the network plan that overlays them — and the resolution logic that turns a name into the full identity the rest of the Azure stack needs. It holds the source-of-truth configuration; the templating module (Catzc.Azure.Templates) consumes it.

Domains

Domain	Area	Name
domain:1	model	Identity and topology model
domain:2	subscription	Subscription resolution
domain:3	environment	Environment resolution
domain:4	catalog	Global catalog accessors

domain:1 — Identity and topology model

The declared shape of the Azure estate: the tenants, the customers (each with a readable key and a short code), the environments (each with a region, region code, and short code), and the subscriptions (each bound to a tenant, optionally to a customer, and serving a set of environments). Overlaid on this is the per-environment IP plan. This is data, not behaviour — it lives in two configuration assets, `azure.yml` and `network.yml`, and is the single source of truth the whole Azure stack resolves against; a quality gate keeps the IP plan in step with the environments. See data-model and network-model.

domain:2 — Subscription resolution

Turning a subscription name into its full identity: its id and tenant, the customer it belongs to (if any), and its per-subscription identity environment. A subscription is the deployment target, and this domain is where "which subscription, and what does that imply" is answered.

domain:3 — Environment resolution

Turning an environment name into its full identity in the context of a serving subscription: its region and region code, its short code, and the resolved subscription that serves it. This is the join between the environment dimension and the subscription dimension.

domain:4 — Global catalog accessors

Whole-estate lookups that aren't tied to one subscription or environment: the list of customers, and the minimum Bicep CLI version the estate requires. These are thin readers over the identity model for callers that need a catalog rather than a single resolved identity.

What the module does

This module is pure resolution over a declared model. It performs no Azure API calls and changes no state — it reads `azure.yml` and `network.yml`, validates each through its config type on load, checks with a quality gate that the network plan and the environments stay in sync, and answers questions about identity. The governing principle is "optimise for ease of configuration and codebase-as-source-of-truth": the configuration is shaped for a human to read and edit, and the non-trivial joins between its dimensions live here, in code, rather than being denormalised into the config.

The model has two independent dimensions — subscriptions and environments — that meet at deployment time. Domain 2 resolves the subscription dimension, domain 3 resolves the environment dimension *in the context of* a subscription, and the catalog accessors in domain 4 expose the whole sets when a caller needs to enumerate rather than resolve. The customer is never a dimension a caller selects directly; it is derived from the resolved subscription, which is why "which customer" is answered through subscription resolution rather than as its own input.

Everything downstream — resource naming, build, deploy — phrases its questions in terms this module answers, so the identity model has exactly one home. Adding a tenant, customer, environment, or subscription is an edit to `azure.yml` (and a matching `network.yml` entry for a standard environment); the validation on load is what stops the two assets from drifting apart.

Division

The module's public functions and configuration, sorted into the domains above.

Domain	Function
domain:1 — Identity and topology model	`Test-AzureNetworkPlan`
config	`azure.yml`
	`network.yml`
domain:2 — Subscription resolution	`Get-AzureSubscription`
	`Get-AzureSubscriptionCustomer`
	`Get-AzureSubscriptionEnvironment`
domain:3 — Environment resolution	`Get-AzureEnvironment`
domain:4 — Global catalog accessors	`Get-AzureCustomers`
	`Get-AzureBicepMinVersion`